

Clearly, if D'_i contains both vertices of Y , D_i contains vertices on either halfspace of H_k in C_0 . Since $D_i \cap C_0$ is convex in C_0 , D_i contains an edge dual to H_k , thus $D'_i \cap C = C$ is the edge $\bar{e}$ dual to H_k in Y : the intersection is convex in C . Otherwise, D'_i contains only one vertex of Y , and the intersection $D'_i \cap C$ is this vertex, which is convex in C . Moreover, in all cases, $\iota_0 \circ c(D'_i) = \iota_0 \circ c_0(D_i) = \mathbb{S}_{\Delta_i}$. This proves that D'_i , c and ι_0 satisfy Assumptions (a) and (b) of Proposition 5.6, (1.).

Since $H_k \notin \mathfrak{K}$, edges of X dual to H_k are not in C_k . Choose such an edge e and p a geodesic path in C_k with the same endpoints. The projection of p in Y joins the two vertices of Y , thus some edge of p does not project to a loop in Y . Let $K_k \in \mathfrak{K}$ be a hyperplane dual to such an edge. By Lemma 2.15, every hyperplane of X transverse to H_k is transverse to K_k . In particular, H_k and K_k are not transverse. Symmetrically, choose an edge f in X dual to K_k and q a geodesic path in C_0 with the same endpoints. As before, some edge of q does not project to a loop in Y , thus has its endpoints in opposite halfspaces of H_k in C_0 . This edge of q must then be dual to H_k . By Lemma 2.15 again, every hyperplane of X transverse to K_k is transverse to H_k .

By specialness of Y , there is a single edge $\bar{f}$ dual to K_k in Y , and every cube spanned by the edge dual to H_k and other edges corresponds to a unique cube spanned by the edge dual to K_k and other edges and vice-versa. Therefore, there exists an involutive combinatorial automorphism α of Y that exchanges $\bar{e}$ and $\bar{f}$ while fixing both vertices and all the other oriented edges. This proves that the collapse $c': Y \rightarrow S_1$ of K_k in Y is a homotopy equivalence. Besides, α induces an isomorphism $\bar{\alpha}: S_0 \rightarrow S_1$ in the sense that $c' \circ \alpha = \bar{\alpha} \circ c$. In particular, $\iota_1 = \iota_0 \circ \bar{\alpha}^{-1}: S_1 \rightarrow S_0 \rightarrow \mathbb{S}$ is a combinatorial isomorphism, and $\iota_1 \circ c' = \iota_0 \circ c \circ \alpha^{-1}$. Moreover, letting $c_1: X \rightarrow S_1$ be the collapse of $\{H_1, \dots, H_{k-1}, K_k\}$, c_1 is a composition $c' \circ d$ of homotopy equivalences, hence a homotopy equivalence itself.

Recall that $\bar{e}$ and $\bar{f}$ are the only edges of Y dual to H_k and K_k respectively. Assume $\bar{e}$ is in D'_i . Then some edge e' parallel to e is in D_i . Let p' be a geodesic path in C_k joining the endpoints of e' . By convexity of $D_i \cap C_k$, p' is contained in D_i . By Corollary 2.16, every edge of p is parallel to an edge of p' . In particular, D_i contains some edge dual to K_k . In Y , its image D'_i then contains $\bar{f}$, the only edge dual to K_k . Symmetrically, if $\bar{f}$ is in D'_i then so is $\bar{e}$. Therefore, the 1-skeleton of D'_i is α -invariant, and by local convexity of D'_i , the whole subcomplex D'_i is α -invariant. In particular, $\iota_1 \circ c'(D'_i) = \iota_0 \circ c(D'_i) = \mathbb{S}_{\Delta_i}$. Moreover, if D'_i contains both vertices of Y , D'_i contains $\bar{e}$, hence contains $\bar{f}$ as well. Thus, the intersection of D'_i with the vertex preimage $C' = \bar{f}$ of c' is convex in C' . This proves that D'_i , c' and ι_1 satisfy Assumptions (a') and (b') of Proposition 5.6, (1.).

By Proposition 5.6, $[(\iota_0)_* \circ (c_0)_* \circ ((\iota_1)_* \circ (c_1)_*)^{-1}] = [(\iota_0)_* \circ (c)_* \circ ((\iota_1)_* \circ (c')_*)^{-1}] \in \mathcal{U}(A_\Gamma; \mathcal{G})$. Thus $[(\iota_1)_* \circ (c_1)_* \circ ((\iota_k)_* \circ (c_k)_*)^{-1}] \in \mathcal{U}(A_\Gamma; \mathcal{G})$ as well. Both collapses c_1 , c_k factor through the collapse of $\{K_k\}$ and its range X' . The induction now proceeds in X' . $\square$

Lemma 5.8. *Let X be a spatial cube complex, and let H_a, H_b, H_c, H_d be four distinct hyperplanes of X . Assume that both collapses $c_{ab}: X \rightarrow S_{ab}$ of $\{H_a, H_b\}$ and $c_{cd}: X \rightarrow S_{cd}$ of $\{H_c, H_d\}$ are homotopy equivalences with range isomorphic to $\mathbb{S}$. Then, up to exchanging H_a and H_b , both collapses $c_{ac}: X \rightarrow S_{ac}$ of $\{H_a, H_c\}$ and $c_{bd}: X \rightarrow S_{bd}$ of $\{H_b, H_d\}$ are homotopy equivalences with range isomorphic to $\mathbb{S}$.*